

Assignment 11.

$$1. \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{N-1} \end{pmatrix} = \frac{1}{\sqrt{N}} \begin{pmatrix} W^{-0.0} & & & W^{-0.(N-1)} \\ & \ddots & & \\ & & \ddots & \\ W^{-(N-1).0} & & & W^{-(N-1)(N-1)} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{pmatrix}$$

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} W^{-kj} x_j = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-i2\pi kj/N} x_j$$

$$= \frac{1}{\sqrt{N}} \begin{bmatrix} e^{-i2\pi(0.0)/N} & e^{-i2\pi(0.1)/N} & \dots & e^{-i2\pi(0.(N-1))/N} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-i2\pi(N-1).0/N} & \dots & \dots & e^{-i2\pi(N-1)(N-1)/N} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{bmatrix} \quad \text{--- ②}$$

$e^{i2\pi m/N} = W^m$, so it equals to ① after substitution.

$$= \frac{1}{\sqrt{N}} \begin{bmatrix} W^{-0.0} & W^{-0.1} & \dots & W^{-0.(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ W^{-(N-1).0} & \dots & \dots & W^{-(N-1)(N-1)} \end{bmatrix} \begin{bmatrix} x_0 \\ \vdots \\ x_{N-1} \end{bmatrix}$$

2. Yes U_{QFT} is Hermitian

$$U_{QFT} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \text{ equal to H gate.}$$

$$U_{QFT}^\dagger = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$U_{QFT}^\dagger = U_{QFT}$ so, it satisfies Hermitian requirements.

$$3. \langle x | = \frac{1}{\sqrt{N}} [W^{-x.0} \quad W^{-x.1} \quad W^{-x.2} \quad \dots \quad W^{-x.(N-1)}]^*$$

$$\langle y | = \frac{1}{\sqrt{N}} [W^{-y.0} \quad W^{-y.1} \quad W^{-y.2} \quad \dots \quad W^{-y.(N-1)}]^*$$

$$\begin{aligned} \langle x_j | y_l \rangle &= \frac{1}{N} \sum_{k=0}^{N-1} W^{-jk} \overline{W^{-lk}} = \frac{1}{N} \sum_{k=0}^{N-1} W^{-jk} W^{lk} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} W^{(l-j)k} \end{aligned}$$

$$\text{if } j=l : \langle x_j | y_j \rangle = \frac{1}{N} \sum_{k=0}^{N-1} 1 = 1$$

if $j \neq l$: let $m = l-j \neq 0 \pmod{N}$ then

$$\sum_{k=0}^{N-1} W^{mk} = \frac{1-W^{mN}}{1-W^m} = \frac{1-1}{1-W^m} = 0$$

$$\langle x_j | y_l \rangle = 0$$

So the rows are orthonormal and $U_{QFT} U_{QFT}^\dagger = I$; U_{QFT} is unitary.

4. For 3 qubit matrix $N = 2^3 = 8$

$$U_{QFT} = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & w^{-1} & w^{-2} & w^{-3} & w^{-4} & w^{-5} & w^{-6} & w^{-7} \\ 1 & w^{-2} & w^{-4} & w^{-6} & w^{-8} & w^{-10} & w^{-12} & w^{-14} \\ 1 & w^{-3} & w^{-6} & w^{-9} & w^{-12} & w^{-15} & w^{-18} & w^{-21} \\ 1 & w^{-4} & w^{-8} & w^{-12} & w^{-16} & w^{-20} & w^{-24} & w^{-28} \\ 1 & w^{-5} & w^{-10} & w^{-15} & w^{-20} & w^{-25} & w^{-30} & w^{-35} \\ 1 & w^{-6} & w^{-12} & w^{-18} & w^{-24} & w^{-30} & w^{-36} & w^{-42} \\ 1 & w^{-7} & w^{-14} & w^{-21} & w^{-28} & w^{-35} & w^{-42} & w^{-49} \end{bmatrix}$$

Given $w = e^{i2\pi/N} = e^{i2\pi/8}$

$$e^{i\pi/4} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i = \frac{1+i}{\sqrt{2}}$$

$$w^8 = (e^{i2\pi/8})^8 = 1, \quad w^{16} = (e^{i2\pi/8})^{16} = 1 = w^{24}$$

5. $U_{QFT} = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & w^1 & w^2 & \dots & w^{(N-1)} \\ 1 & w^1 & w^2 & \dots & w^{(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^1 & w^2 & \dots & w^{(N-1)} \end{pmatrix}_{(N \times N)}$

$$U_{IQFT} = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & w^1 & w^2 & \dots & w^{(N-1)} \\ 1 & w^1 & w^2 & \dots & w^{(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^1 & w^2 & \dots & w^{(N-1)} \end{pmatrix}$$

$$U_{QFT} \cdot U_{IQFT} = \frac{1}{N} \begin{bmatrix} N & 0 & 0 & 0 \\ 0 & N & 0 & 0 \\ 0 & 0 & N & 0 \\ 0 & 0 & 0 & N \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

6. $U_{IQFT} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ w^1 & w^1 & w^2 & \dots & w^{(N-1)} \\ w^2 & w^2 & w^4 & \dots & w^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ w^{(N-1)} & w^{(N-1)} & w^{2(N-1)} & \dots & w^{(N-1)(N-1)} \end{bmatrix}$

For 2 qubit $w = i$

$$U_{QFT} = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -i \\ 1 & i & -1 & -i \end{bmatrix}$$

$$= \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -i \\ 1 & i & -1 & -i \end{bmatrix}$$

$$U_{QFT} \cdot U_{IQFT} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -i \\ 1 & i & -1 & -i \end{pmatrix} \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -i \\ 1 & i & -1 & -i \end{pmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

8. As we know when apply H gate to a qubit, it will be $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

So, QFT = Hadamard gate for 1 qubit and we know there will be no phase shift if all bits are zero then we know they will equal probability in all states if all the given inputs are in starting state $|0\rangle$.

9. $U_{\text{CPS}, \phi} U_{\text{CPS}, -\phi} = I$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{-i\phi} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi - i\phi} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

10. $U_{\text{swap}} = U_{\text{swap}}^\dagger \Rightarrow U_{\text{swap}} \cdot U_{\text{swap}} = I$

1st method $U_{\text{swap}}^\dagger = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^\dagger = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

2nd method $U_{\text{swap}} U_{\text{swap}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I$

12. We need $n-1$ $U_{\text{CPS}, \phi}$ gates with ϕ from $-\frac{2\pi}{2^2}$ to $-\frac{2\pi}{2^n}$

Hadamard gates : n gates

Controlled Rotation gates :

1st qubit $(n-1)$ gates

2nd qubit $(n-2)$ gates

⋮

the last qubit : 0 gate

total controlled rotation gates : $(n-1) + (n-2) + \dots + 1 = \frac{n(n-1)}{2}$

Swap gate $\lfloor n/2 \rfloor$ gates

total execution time $n + \frac{n(n-1)}{2} + \lfloor n/2 \rfloor$ units

$n + \frac{n^2 - n}{2} + \lfloor n/2 \rfloor \approx O(n^2)$ n is the number of qubits.

Eg. 2 qubits : $2 + \frac{2^2 - 2}{2} + \lfloor 2/2 \rfloor = 2 + 1 + 1 = 4$ units.

14. $|1001100\rangle \xrightarrow{\text{SWAP}} |0011001\rangle$

15. A single qubit state with a global phase $e^{i\alpha}$ is:

$$|\psi'\rangle = e^{i\alpha} |\psi\rangle = e^{i\alpha} (a|0\rangle + b|1\rangle)$$

The expectation value of $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is:

$$\begin{aligned} \langle \psi' | \sigma_z | \psi' \rangle &= \langle \psi | e^{-i\alpha} \sigma_z e^{i\alpha} | \psi \rangle = e^{-i\alpha} e^{i\alpha} \langle \psi | \sigma_z | \psi \rangle \\ &= \langle \psi | \sigma_z | \psi \rangle \end{aligned}$$

Since $e^{-i\alpha} e^{i\alpha} = 1$, the global phase cancels exactly.

$$\text{The expectation value is: } \langle \sigma_z \rangle = |a|^2 - |b|^2$$

which is independent of α . The global phase has no physical effect.

16. The general state on the Bloch sphere is:

$$|\psi\rangle = \cos\frac{\theta}{2} |0\rangle + e^{i\phi} \sin\frac{\theta}{2} |1\rangle$$

with $\langle \sigma_z \rangle = \cos\theta$, $\langle \sigma_x \rangle = \sin\theta \cos\phi$, $\langle \sigma_y \rangle = \sin\theta \sin\phi$.

$|-\rangle$ ($\theta = \pi/2$, $\phi = \pi$): $\langle \sigma_x \rangle = \sin(\pi/2) \cos\pi = -1$, $\langle \sigma_y \rangle = 0$, $\langle \sigma_z \rangle = 0$.
This is the $-x$ pole.

$$|-\rangle = \cos\frac{\pi}{4} |0\rangle + e^{i\pi} \sin\frac{\pi}{4} |1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$|+\rangle$ ($\theta = \pi/2$, $\phi = \pi/2$): $\langle \sigma_y \rangle = \sin(\pi/2) \sin(\pi/2) = +1$, $\langle \sigma_x \rangle = 0$, $\langle \sigma_z \rangle = 0$.
This is the $+y$ pole.

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$|-\rangle$ ($\theta = \pi/2$, $\phi = 3\pi/2$): $\langle \sigma_y \rangle = \sin(\pi/2) \sin(3\pi/2) = -1$, $\langle \sigma_x \rangle = 0$, $\langle \sigma_z \rangle = 0$.
This is the $-y$ pole.

$$|-\rangle = \frac{1}{2} (|0\rangle - i|1\rangle)$$

17. Using $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$.

Eq (1): $\langle\sigma_x\rangle = \sin\theta\cos\phi$

Since $\sigma_x|0\rangle = |1\rangle$ and $\sigma_x|1\rangle = |0\rangle$:

$$\begin{aligned}\langle\sigma_x\rangle &= (\cos\frac{\theta}{2}\langle 0| + e^{-i\phi}\sin\frac{\theta}{2}\langle 1|)(\cos\frac{\theta}{2}|1\rangle + e^{i\phi}\sin\frac{\theta}{2}|0\rangle) \\ &= e^{i\phi}\cos\frac{\theta}{2}\sin\frac{\theta}{2} + e^{-i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2} \\ &= \cos\frac{\theta}{2}\sin\frac{\theta}{2}(e^{i\phi} + e^{-i\phi}) = \frac{\sin\theta}{2} \cdot 2\cos\phi = \sin\theta\cos\phi\end{aligned}$$

Eq (2): $\langle\sigma_y\rangle = \sin\theta\sin\phi$

Since $\sigma_y|0\rangle = i|1\rangle$ and $\sigma_y|1\rangle = -i|0\rangle$:

$$\begin{aligned}\langle\sigma_y\rangle &= (\cos\frac{\theta}{2}\langle 0| + e^{-i\phi}\sin\frac{\theta}{2}\langle 1|)(i\cos\frac{\theta}{2}|1\rangle - ie^{i\phi}\sin\frac{\theta}{2}|0\rangle) \\ &= -ie^{i\phi}\cos\frac{\theta}{2}\sin\frac{\theta}{2} + ie^{-i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2} \\ &= i\cos\frac{\theta}{2}\sin\frac{\theta}{2}(e^{-i\phi} - e^{i\phi}) = i \cdot \frac{\sin\theta}{2} \cdot (-2i\sin\phi) \\ &= \sin\theta\sin\phi\end{aligned}$$

18. Step 1: Verify normalization:

$$|\alpha|^2 = \frac{9}{16} + \frac{3}{16} = \frac{12}{16} = \frac{3}{4}, \quad |\beta|^2 = \frac{3}{16} + \frac{1}{16} = \frac{4}{16} = \frac{1}{4} \rightarrow |\alpha|^2 + |\beta|^2 = 1$$

Step 2: Extract global phase:

$$\alpha = \frac{\sqrt{3}}{2}(\frac{\sqrt{3}}{2} - \frac{i}{2}) = \frac{\sqrt{3}}{2}e^{-i\pi/6}, \quad \beta = \frac{1}{2}(\frac{\sqrt{3}}{2} - \frac{i}{2}) = \frac{1}{2}e^{-i\pi/6}$$

$$|\psi\rangle = e^{-i\pi/6} [\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle]$$

The global phase $e^{-i\pi/6}$ is physically irrelevant. Comparing with $\cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$

$$\cos\frac{\theta}{2} = \frac{\sqrt{3}}{2} \rightarrow \frac{\theta}{2} = \frac{\pi}{6} \rightarrow \theta = \frac{\pi}{3}$$

$$e^{i\phi}\sin\frac{\theta}{2} = \frac{1}{2} \in \mathbb{R}^+ \rightarrow \phi = 0$$

$$\text{Bloch vector } (x, y, z) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta) = (\frac{\sqrt{3}}{2}, 0, \frac{1}{2})$$

The state lies on the xz -plane, tilted 60° from the north pole toward the x -axis

19. $U(\theta, \phi, \lambda) = \begin{pmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\phi+\lambda)} \cos \frac{\theta}{2} \end{pmatrix}$

Substituting $\theta = \pi/2$, $\phi = 0$, $\lambda = \pi$:

$$U(\pi/2, 0, \pi) = \begin{pmatrix} \cos \frac{\pi}{4} & -e^{i\pi} \sin \frac{\pi}{4} \\ e^{i0} \sin \frac{\pi}{4} & e^{i(\pi)} \cos \frac{\pi}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = H$$

20. The Hadamard gate acts as a rotation by π about the $(x+z)/\sqrt{2}$ axis, swapping the x - and z -axis

Case 1: $|+\rangle \xrightarrow{H} |0\rangle$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \xrightarrow{H} \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle) = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) = |0\rangle$$

On the Bloch sphere: $|+\rangle$ sits at $(1, 0, 0)$ on the $+x$ axis.

After H it moves to $(0, 0, 1)$, the north pole $|0\rangle$.

Case 2: $|1\rangle \xrightarrow{H} |-\rangle$

$$|1\rangle \xrightarrow{H} H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

On the Bloch sphere: $|1\rangle$ sits at $(0, 0, -1)$ on the $-z$ axis.

After H it moves to $(-1, 0, 0)$, the $-x$ axis, i.e. $|-\rangle$.

In both cases, the Hadamard effectively reflects the Bloch vector through the plane defined by the $(x+z)/\sqrt{2}$ axis, exchanging $x \leftrightarrow z$ coordinates.